BLOODLINK: SMART IOT-BASED BLOOD STORAGE RACK FOR INVENTORY AND COLD CHAIN MONITORING

A MINI PROJECT REPORT

submitted by

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for the award of the degree of

BACHELOR OF TECHNOLOGY



DEPARTMENT OF ELECTRONICS ENGINEERING
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CERTIFICATE

This is to certify that the report entitled **BloodLink: Smart IoT-Based Blood Storage Rack for Inventory and Cold Chain Monitoring** submitted by Abhijith Santhosh (MGP23ECE001), Aswin S (MGP23ECE018), Elias Janson (MGP23ECE022), Johan Paul (MGP23CS082) to SAINTGITS College of Engineering (Autonomous), affiliated to the APJ Abdul Kalam Technological University, Thiruvananthapuram, in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Electronics and Computer Engineering, is a bonafide record of project work done by them under my supervision. This report in any form has not been submitted to any other University or Institute for any purpose.

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DECLARATION

We undersigned hereby declare that the Mini Project report titled "**BloodLink: Smart IoT-Based Blood Storage Rack for Inventory and Cold Chain Monitoring**" submitted for partial fulfillment of the requirements for the award of the degree of Bachelor of Technology of SAINTGITS College of Engineering (Autonomous) affiliated to the APJ Abdul Kalam Technological University, Thiruvananthapuram is a bonafide work carried out by us under the guidance of Er. Anish M George, Assistant Professor, Department of Electronics Engineering. This submission represents our ideas in our own words and where ideas or words of others have been included, we have adequately and accurately cited and referenced the original sources. We also declare that we have adhered to ethics of academic honesty and integrity and have not misrepresented or fabricated any data or idea or fact or source in our submission. We understand that any violation of the above will be a cause for disciplinary action by the institute and/or the University and can also evoke penal action from the sources which have thus not been properly cited or from whom proper permission has not been obtained. This report has not been previously formed the basis for the award of any degree, diploma or similar title of any other University.

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ABSTRACT

This project presents the **Design and Implementation of BloodLink, an IoT-Based Smart Blood Storage Rack for Inventory and Storage-Condition Monitoring**. Manual recording of blood inventory causes delayed updates, stock errors, difficulty in locating the required blood group, and unnoticed expiry, which lead to avoidable wastage of a scarce and perishable resource.

BloodLink addresses this with a low-cost, 3D-printed prototype rack that combines Radio Frequency Identification (RFID), embedded sensing, a cloud backend, and a web dashboard. Each blood bag carries an RFID tag with a unique identifier. An ESP8266-based node with an RFID reader and an OLED display identifies the bag when it is scanned and gives local feedback, while a second ESP8266-based node monitors slot occupancy using infrared sensors and records the storage temperature. Both nodes send data over Wi-Fi to a Supabase backend, where an Edge Function receives the data and a PostgreSQL database stores bag records, slot status, device heartbeats, temperature readings, and activity events. A web dashboard hosted on Vercel presents the inventory, storage slots, alerts, activity log, and device connectivity.

The prototype demonstrates a four-slot storage model in which each slot can hold multiple individually identified bags; the two RFID tags used in the demonstration do not limit the database capacity. The dashboard outputs confirm that scans, heartbeats, slot status, and temperature readings are recorded and displayed correctly. The system monitors environmental conditions only, and RFID identifies a tag and not the medical suitability of a bag. The work is therefore a proof of concept and a foundation for validated temperature thresholds, expiry alerts, stronger access control, and multi-bank deployment.

KEYWORDS: Internet of Things; RFID; ESP8266; Blood Bank Inventory; Cold Chain Monitoring; Supabase; Web Dashboard; Embedded Systems.

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CHAPTER 1

INTRODUCTION

This project develops BloodLink, an IoT-based smart blood storage rack that identifies blood bags using RFID, tracks their storage slots, monitors storage conditions, and presents the information on a cloud-connected web dashboard.

1.1 General

Blood is a perishable and life-saving resource that cannot be manufactured and must be collected from donors, tested, processed, stored under controlled conditions, and issued before it expires. Every blood bank therefore needs to know, at any moment, how many bags of each blood group it holds, where each bag is located, when it expires, and whether the conditions in which it is kept are acceptable. Accurate and timely information on these points directly affects how quickly a hospital can be supplied and how much blood is lost to wastage.

In many blood banks, inventory is still recorded manually in registers or spreadsheets. Manual recording is slow and prone to human error, particularly during busy periods and emergencies. Stock counts may lag behind the physical stock, a bag may be placed in the wrong compartment, and bags nearing their expiry date may remain unnoticed until they have to be discarded. Storage temperature is usually checked at intervals, so a deviation that occurs between two checks may not be noticed in time.

Radio Frequency Identification (RFID) offers a practical way to remove much of this manual effort. An RFID tag attached to a blood bag carries a unique identifier that a reader can read within a short range without line of sight. When this identifier is linked to a database record, every scan becomes a reliable and time-stamped inventory event. Combined with low-cost embedded sensors and cloud connectivity, RFID can provide near real-time visibility of the stock without requiring expensive infrastructure.

This project builds a compact prototype that integrates these ideas. Each blood bag is registered against a unique RFID tag. An ESP8266-based controller reads the tag, shows local feedback on an OLED display, and sends the event over Wi-Fi to a Supabase cloud backend. A second controller monitors storage-slot occupancy and temperature. A web dashboard shows the stored bags, the state of the storage slots, alerts, device connectivity, and the history of recent activity. The prototype is intentionally low-cost so that the approach can be understood, demonstrated, and extended by students and small blood-storage facilities.

It is important to state the role of the system clearly. BloodLink is an information and monitoring prototype. RFID identifies a tag; it does not independently verify the blood group, the expiry date, or the medical suitability of the bag. These details come from the record registered for the bag and must be entered and validated correctly by authorised staff. Similarly, the temperature readings produced by the prototype are environmental readings, and they do not establish that blood is safe for transfusion.

1.2 Objectives

The objectives of the project are to design, build, and demonstrate a prototype that supports accurate blood inventory management and storage monitoring. The main objectives are as follows:

RFID-Based Identification: To identify individual blood bags automatically using RFID tags that carry unique identifiers, so that each bag has its own traceable record.

Slot-Based Inventory Organisation: To organise the inventory using four storage slots associated with different blood groups, with the ability to hold multiple bags in each slot, and to record the slot location of every bag.

Environmental Monitoring: To monitor the storage temperature (and humidity, where the sensor supports it) using connected sensors, and to detect whether a storage slot is occupied using infrared sensors.

Cloud Connectivity: To transmit device and sensor information over Wi-Fi to

a Supabase backend, and to store it in a database that maintains bag records, slot status, temperature readings, device heartbeats, and activity events.

Local Feedback: To give the operator immediate feedback on scanning through an OLED display on the rack.

Web Dashboard: To provide a web dashboard that presents inventory, storage slots, alerts, the activity log, and device connectivity.

Reliable Time-Stamping: To maintain blood-bag records in which the `created_at` field of the bag record is the agreed source of the time at which the bag entered the system.

Scalable Data Model: To demonstrate the workflow using two RFID tags while keeping the database design capable of holding many more bags and slots.

Foundation for Alerts and Traceability: To provide a base on which alerts for abnormal temperature and approaching expiry, stronger traceability, and multi-site monitoring can be built in future.

1.3 Scope of the Project

This project covers the design, development, and demonstration of a prototype smart blood storage rack. The key areas covered are described below.

Application Scenario: The system is intended for small blood storage facilities, hospital blood storage units, and educational demonstrations of RFID-based inventory and monitoring. It is a prototype and has not been tested or validated in an operating blood bank.

Technology Domain: The project belongs to the domains of the Internet of Things, embedded systems, and healthcare automation. It combines RFID identification, microcontroller programming, sensor interfacing, cloud databases, and web development.

System Architecture: The system follows a three-layer architecture. The device layer consists of two ESP8266 nodes, sensors, and the RFID reader. The cloud layer consists of a Supabase Edge Function and a PostgreSQL database. The

application layer is a web dashboard hosted on Vercel.

Blood-Bag Identification: Each blood bag is represented by a record with a unique RFID tag identifier. The demonstration uses two tags, but the data model is not limited to two bags, and each slot may hold several bags.

Four-Slot Storage Model: The physical model has four storage slots, each associated with a different blood group. The slots are identified in the database as slots of a rack (for example, `RACK-1` with slots `S1-01`, `S1-02`, `S2-01`, and `S2-02`).

Storage Monitoring: The system records the temperature of the storage environment and the occupancy state of slots that have an infrared sensor assigned. The acceptable temperature range is kept as a configurable parameter and has to be set according to the storage requirements of the blood component and the applicable guidelines.

Dashboard and Activity Records: The dashboard presents an overview of the inventory, the blood inventory table, storage slots, open alerts, recent activity, and the connectivity of the devices based on the last heartbeat received.

Cost and Accessibility: The prototype uses low-cost and readily available components and a 3D-printed enclosure, so that the total cost remains within a student project budget.

Limitations: The prototype does not include refrigeration, does not verify the medical validity of a bag, has not undergone clinical validation, and does not claim regulatory approval or medical-grade performance.

Future Scope: The design can be extended with validated temperature thresholds, automatic expiry alerts, stricter authentication and Row Level Security, a larger rack, and support for several blood banks.

CHAPTER 2

LITERATURE REVIEW

A review of existing literature was carried out to understand how RFID, IoT, and information systems have been applied to blood-bank inventory management and storage monitoring. Five papers published between 2022 and 2025 were surveyed in the final review, and one earlier paper on RFID in blood bank supply chains was added to give background. Each work was compared on its identification method, monitoring scope, and limitations. The key papers are summarised in Table 2.1.

Table 2.1: Consolidation of Literature Review

Sl.	**Title**	**Authors & Publication**	**Technique Used and Observations**
1	Radiofrequency Identification Tracking System (RFID) Significantly Improves Blood Bank Inventory Management and Decreases Staff Work Effort	A. T. da Souza et al., Transfusion, vol. 64, no. 4, 2024.	**Technique:** RFID tracking of blood bank inventory. **Observation:** Improves stock accuracy and reduces staff work effort. **Gap:** Low-cost storage-rack integration and continuous temperature monitoring are not the main focus.
2	Automating Inventorying of Blood Stations: A System Based on Ultra-High-Frequency (UHF) RFID Technology	Transfusion Clinique et Biologique, vol. 29, no. 2, 2022.	**Technique:** UHF RFID for automatic inventory counts of blood stations. **Observation:** Automates inventorying of stored units. **Gap:** UHF infrastructure is more complex and costly than short-range RFID suited to a low-cost prototype.
3	Design and Practice of an Emergency Blood Allocation System Based on Radiofrequency Identification Technology	S. Xie et al., Vox Sanguinis, vol. 118, no. 11, 2023.	**Technique:** RFID-based allocation of blood during emergencies. **Observation:** Supports faster and more organised allocation. **Gap:** Targets large-scale allocation, not a compact storage rack with environmental monitoring.

Sl.	Title	Authors & Publication	Technique Used and Observations
4	RFID in Blood Supply Chains: Research Progress and Developments	P. Li, Y. Chen, S. Yan, Military Medical Sciences, vol. 49, no. 5, 2025.	**Technique:** Review of RFID adoption across blood supply chains. **Observation:** Summarises research progress and developments. **Gap:** A broad review; does not implement a low-cost integrated prototype for rack-level monitoring.
5	Analysis of Temperature Distribution in Blood Banks Through Storage of Measurement Results with IoT Monitoring in the Blood Donation Unit of Indonesian Red Cross Surabaya	F. A. S. Wardhana et al., Indonesian Journal of Electronics, Electromedical Engineering, and Medical Informatics, vol. 5, no. 2, 2023.	**Technique:** IoT monitoring that logs and analyses temperature distribution in blood bank storage. **Observation:** Provides stored measurement data for analysis of storage conditions. **Gap:** Focuses on temperature only; RFID inventory and physical slot identification are not covered.
6	Utilizing Radiofrequency Identification Technology to Improve Safety and Management of Blood Bank Supply Chains	A. Coustasse et al., Telemedicine and e-Health, 2015.	**Technique:** RFID applied to the blood bank supply chain. **Observation:** Discusses the use of RFID to improve safety and management of blood supply chains. **Gap:** Addresses the supply chain as a whole rather than rack-level slot tracking with storage monitoring.

The surveyed work falls into two groups. The first group applies RFID to blood bank inventory and the supply chain [1–4, 6]. These studies show that automatic identification improves the accuracy of stock records and reduces staff effort, and that RFID can support rapid allocation of blood during emergencies. The second group uses IoT to monitor and log storage temperature in blood banks [5]. This work shows the value of continuous measurement as compared with periodic manual checks.

In addition to these papers, the project relies on the technical documentation of the components and platforms used: the MFRC522 RFID reader integrated circuit [7], the ESP8266 microcontroller and its Arduino core [8, 9], the Supabase platform including Edge Functions and Row Level Security [10, 11], and the React

library used for the dashboard [12].

2.1 Gaps in Literature

The review shows that inventory tracking and temperature monitoring are mostly treated as separate problems. RFID-based systems improve inventory accuracy but do not usually report the conditions of storage, whereas IoT temperature monitors log storage conditions but do not identify the individual bags or the slot in which they are kept.

Several of the inventory systems are designed for large-scale deployment, use UHF readers, or are meant for allocation across many sites. Such infrastructure is costly and complex for a small blood bank or a student prototype. Broad review articles describe the potential of RFID but do not present a low-cost implementation that students or small facilities can reproduce.

These gaps show the need for a compact and affordable system in which a short-range RFID reader identifies each blood bag, the physical storage slot of the bag is known, storage conditions are measured, and everything is synchronised to a cloud database and presented on a dashboard. This combination is the focus of the present project.

CHAPTER 3

COMPONENTS AND WORKING METHODOLOGY

3.1 Hardware Components

The following hardware components are used in the development of the BloodLink prototype. Exact module part numbers and pin assignments are defined in the project firmware and wiring and are not repeated here.

Table 3.1: Hardware Components (Part 1)

Sl. No.	Component	Function
1	ESP8266 NodeMCU (two nodes)	Acts as the main controller of each node and provides Wi-Fi communication with the cloud backend. One node is used for RFID scanning and the OLED display (device code ESP8266-RFID-01) and the other for slot and temperature sensing (device code ESP8266-SENSOR-01).
2	RFID reader module (RC522 type)	Reads the RFID tags attached to blood bags. The RC522 module is based on the MFRC522 integrated circuit, which operates at 13.56 MHz and communicates with the microcontroller over a serial peripheral interface.
3	RFID tags / cards	Provide a unique identifier for each blood bag. Two tags (key-fob type tags, as shown in the prototype photographs) are used in the live demonstration; each represents one individual bag.
4	OLED display	Shows the status of the system and scan feedback locally on the rack, so that the operator receives immediate confirmation.

Table 3.2: Hardware Components (Part 2)

Sl. No.	Component	Function
5	Infrared (IR) proximity sensors	Detect whether a storage slot is occupied. In the current configuration the sensors IR-01 and IR-02 are assigned to slots S1-01 and S1-02; the remaining slots are available for further sensors.
6	Temperature / humidity sensor	Measures the environmental conditions of the storage area. The readings are uploaded by the sensor node and shown on the dashboard.
7	Power supply	Powers the ESP8266 nodes and the connected modules from a regulated supply.
8	Wires and breadboard / PCB	Provide the electrical interconnections between the controllers, the reader, the display, and the sensors.
9	Four-slot 3D-printed storage model	Represents separate blood-group storage locations. It has a control panel with an OLED window and RFID reader area, hinged compartment doors, ventilation slots, and an electronics bay. Each slot may hold multiple bags.

Summary: The hardware components selected for the prototype enable identification of blood bags, detection of slot occupancy, measurement of storage conditions, local feedback, and cloud connectivity. All components are low-cost and readily available, and the ESP8266 has built-in Wi-Fi, so no separate network module is needed. The use of two nodes separates the scanning function from the sensing function, which keeps the firmware of each node simple.

3.2 Software Components

The software components used in this project enable device firmware, secure data transfer, storage, and presentation of information.

ESP8266 Firmware (Arduino framework): The firmware running on the ESP8266 nodes initialises the RFID reader, the OLED display, and the sensors, connects to the configured Wi-Fi network, reads RFID tags and sensor values, updates the OLED, and sends the data to the backend over Wi-Fi. The ESP8266 Arduino core is used as the development framework [9].

Supabase Backend: Supabase provides the PostgreSQL database and the server-side environment for the project. The database stores the bag records, storage-slot records, device records, readings, and activity events.

Supabase Edge Function: An Edge Function receives the data sent by the devices, performs the required server-side operations, and records device heartbeats. The dashboard reports device connectivity based on the last heartbeat recorded by the Edge Function.

Web Dashboard (React, hosted on Vercel): The dashboard is a web application built with React [12] and deployed on Vercel. It presents an overview, the blood inventory, storage slots, alerts, the activity log, device connectivity, and settings.

Summary: The software architecture follows a simple division of work. The firmware performs sensing and communication, the Edge Function and database perform validation, storage, and bookkeeping, and the dashboard performs presentation. This division keeps the microcontrollers lightweight, because they do not need to run any application logic apart from reading and sending data.

3.3 Hardware Design Proposal

The hardware design describes the physical implementation of the BloodLink prototype, as shown in Fig. 3.1. The rack is a 3D-printed enclosure with a control panel on the front and a compartment area beside it. The OLED display window is located at the upper left of the front panel, and the RFID reader module is mounted below it at the lower left so that a tag can be presented to the reader easily. Four hinged compartment doors with handles provide access to the four storage slots. Ventilation slots are provided at the bottom of the front panel, and the top of the enclosure is open to give access to the electronics bay that houses the ESP8266 controllers and the wiring.

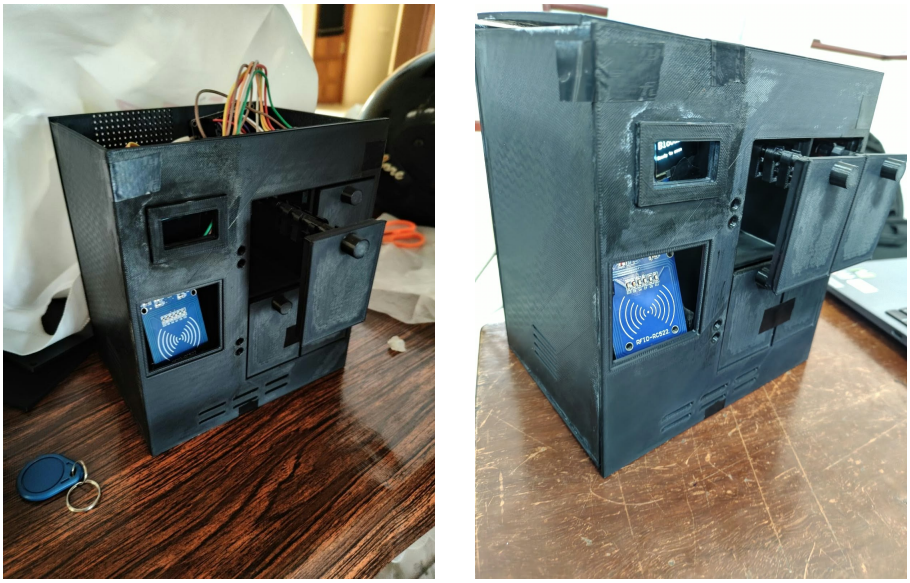


Figure 3.1: Hardware Design of the BloodLink Prototype (front and side-angle views)

3.4 Working Methodology

The system operates through a defined pipeline that integrates identification, sensing, communication, storage, and presentation. The methodology follows the block diagram given in Section 3.5 and has the following stages.

1. **Initialisation:** On start-up each ESP8266 node initialises its peripherals. The RFID node initialises the RFID reader and the OLED display, and the sensor node initialises the infrared sensors and the temperature sensor.
2. **Wi-Fi Connection:** Each node connects to the configured Wi-Fi network and identifies itself to the backend with its device code.
3. **RFID Scanning:** The operator presents the tag of a blood bag to the RFID reader. The reader returns the unique identifier of the tag to the ESP8266.
4. **Association with a Bag Record:** The tag identifier is associated with the corresponding blood-bag record in the database. A tag that has been registered resolves to its bag; a tag that is not registered is not associated with any bag.
5. **Backend Processing:** The identifier is sent to the Edge Function, which looks up or updates the corresponding records according to the implemented API and writes an activity event (an `rfid_scan` event with the device, the bag, and the time).
6. **Local Feedback:** The OLED display shows scanning feedback and the available bag information so that the operator knows the result of the scan.

7. **Environmental and Slot Sensing:** The sensor node reads the temperature (and humidity) and the state of the infrared sensors that indicate slot occupancy.
8. **Data Transmission:** The sensor node sends the readings to the backend over Wi-Fi at regular intervals. Both nodes also send heartbeat messages so that the dashboard can show whether each device is active.
9. **Storage in Supabase:** Supabase stores or updates the appropriate records: the bag entry (using `blood_bags.created_at` as the time of entry), the slot status, the latest temperature reading, the device `last_seen_at`, and the activity event.
10. **Dashboard Presentation:** The web dashboard retrieves the records from the backend and presents the inventory, storage slots, alerts, activity, and device connectivity to the operator.
11. **Repeated Operation:** The process repeats for every further bag that is scanned and for every sensing interval.

Four-Slot Storage Logic: The rack has four storage slots, each associated with a different blood group. Each slot can hold more than one bag, and every bag has its own record and its own RFID tag, so the position of a bag is described by the slot with which it is associated. The two tags available for the demonstration show the process only. The number of demonstration tags does not set the maximum capacity of the inventory, which is determined by the database and the physical size of the slots. The blood group associated with each slot is a configuration of the rack and is not fixed by the design.

Temperature Monitoring Logic: The readings from the temperature sensor are stored with their time and shown on the dashboard. The acceptable temperature range is a configurable parameter. For clinical use it must be derived from the storage requirements of the specific blood component and the applicable guidelines (for example, red-cell components are kept under refrigeration within a narrow range), and a demonstration threshold must not be mistaken for a clinical limit. The sensor measures the environment where it is placed, and its reading must not be taken as the temperature inside every blood bag unless the sensor arrangement is designed and validated for that purpose.

Summary: The working methodology forms a continuous loop in which bags are identified, slot and environmental information is collected, data is sent to the cloud, and the dashboard is updated. Heartbeats make the health of the devices

visible to the operator, and the activity log provides a time-stamped history of scans.

3.5 Block Diagram

The block diagram in Fig. 3.2 shows the overall architecture of the system. It illustrates the flow of data from the inputs (RFID reader, IR slot sensors, and temperature sensor) through the ESP8266 controllers and the Wi-Fi link to the Supabase backend, and finally to the web dashboard presented to the user.

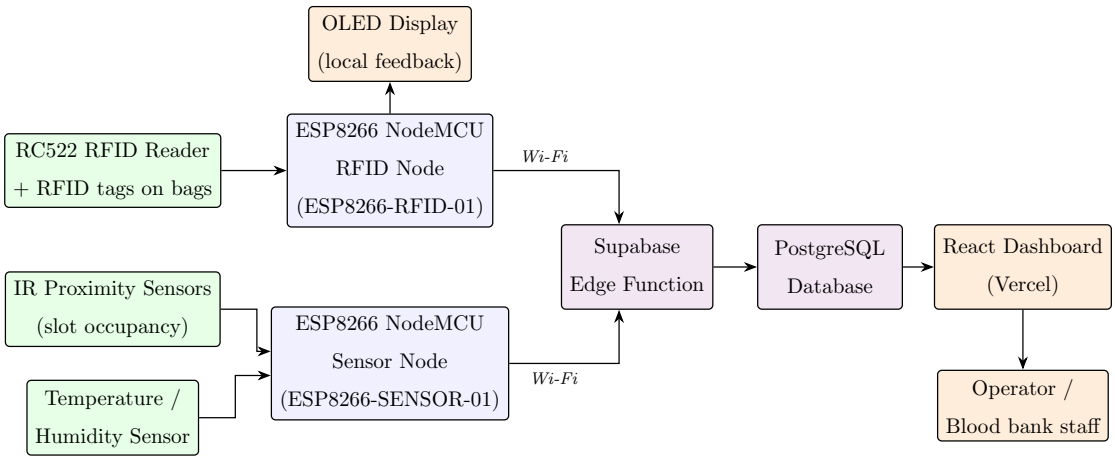


Figure 3.2: Block Diagram of the BloodLink System

3.6 Circuit Diagram

The interconnection diagram in Fig. 3.3 shows how the modules of the prototype are connected at module level. The RFID node connects the RC522 reader through a serial peripheral interface and the OLED display through its serial display interface. The sensor node reads the infrared sensors of the occupied slots and the temperature sensor through their signal lines. Both nodes are powered from the power supply and both reach the cloud through their built-in Wi-Fi.

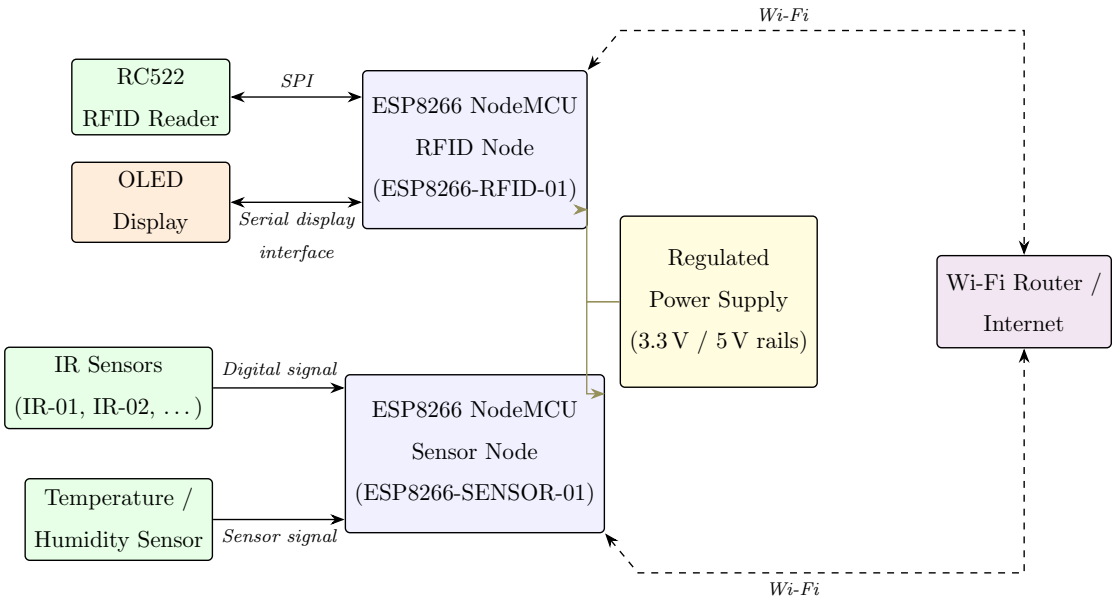


Figure 3.3: Module-Level Interconnection (Circuit) Diagram of the BloodLink Prototype. GPIO pin assignments are defined in the firmware and are not shown.

The diagram is drawn at module level and does not label individual GPIO pins. The signal levels of the ESP8266 and of the RC522 module are 3.3 V, and the power rails supply the modules within their rated voltage. The exact pin mapping is recorded in the firmware configuration of the prototype.

3.7 Database and System Architecture

The system architecture in Fig. 3.4 divides BloodLink into a device layer, a cloud layer, and an application layer. The device layer senses and identifies. The cloud layer receives, validates, and stores. The application layer presents the information to the operator.

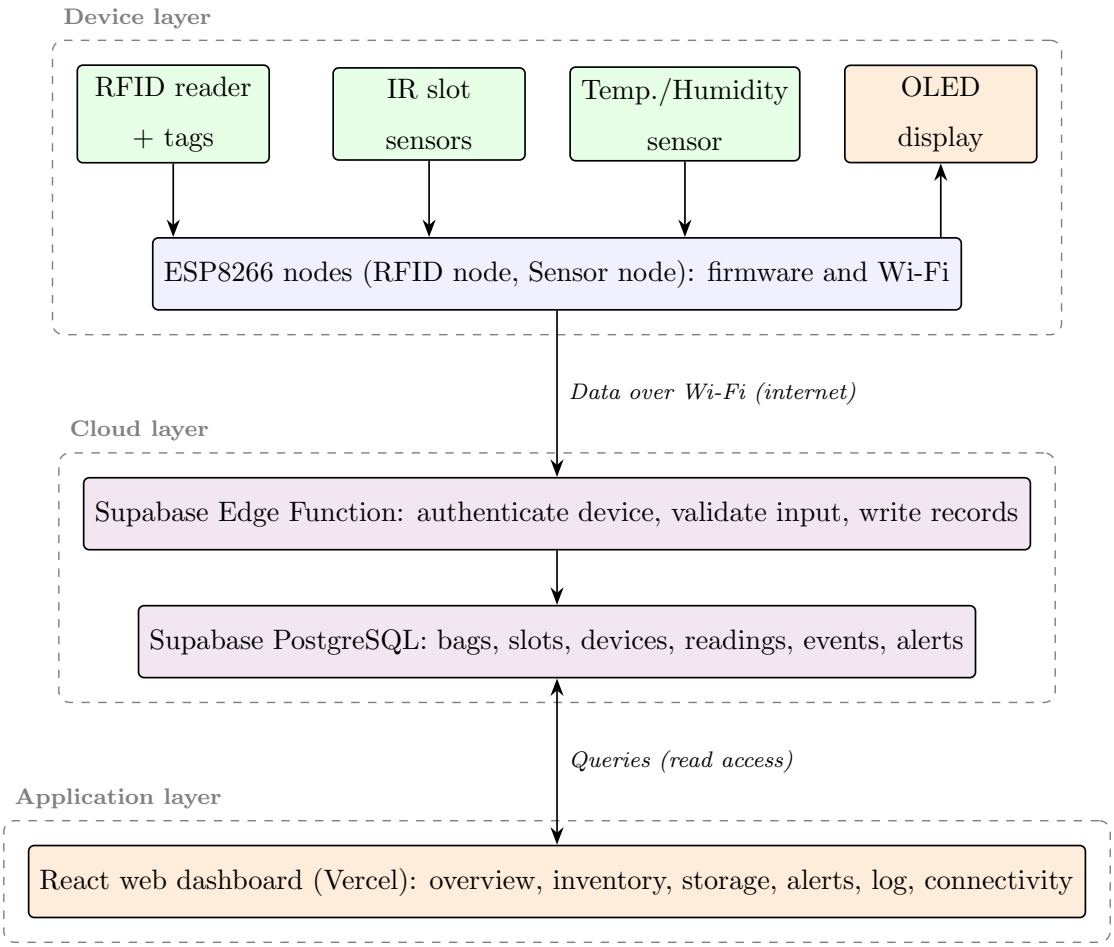


Figure 3.4: System Architecture of BloodLink

RFID-to-Database Data Flow

The data flow of a scan is shown in Fig. 3.5. A tag presented to the reader is read by the ESP8266, which sends its identifier to the backend. If the identifier belongs to a registered bag, the record is processed and an activity event is stored, and the OLED shows the result. Otherwise the scan is reported as an unregistered tag and no bag record is changed.

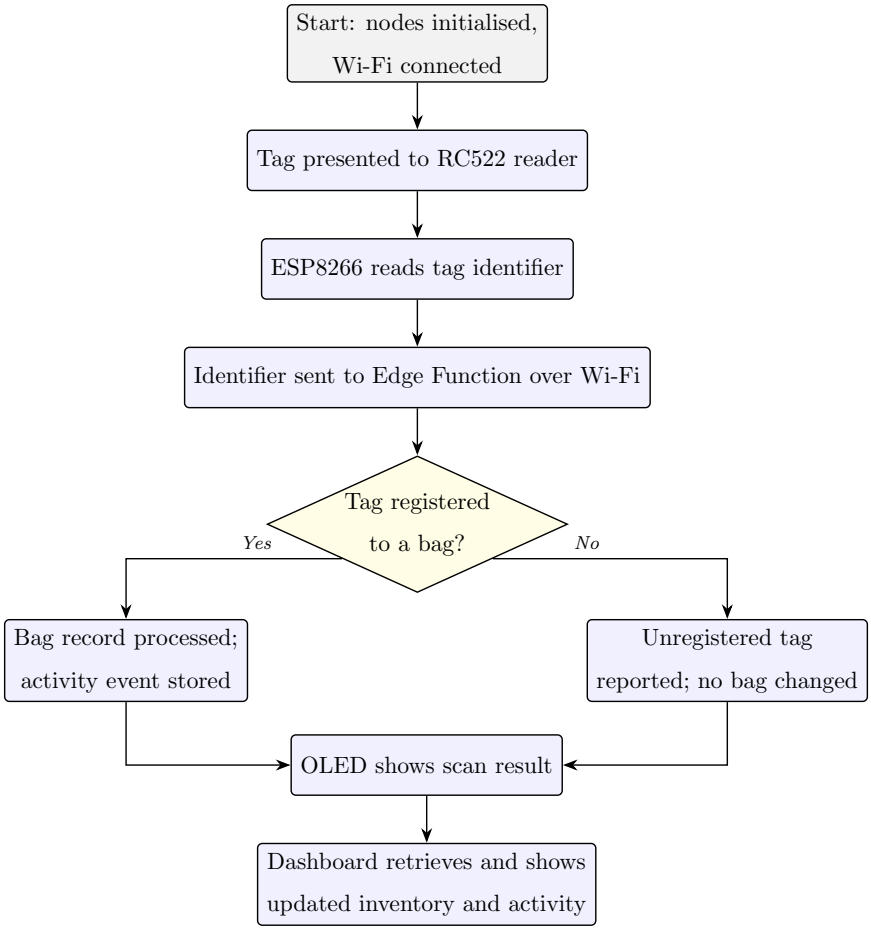
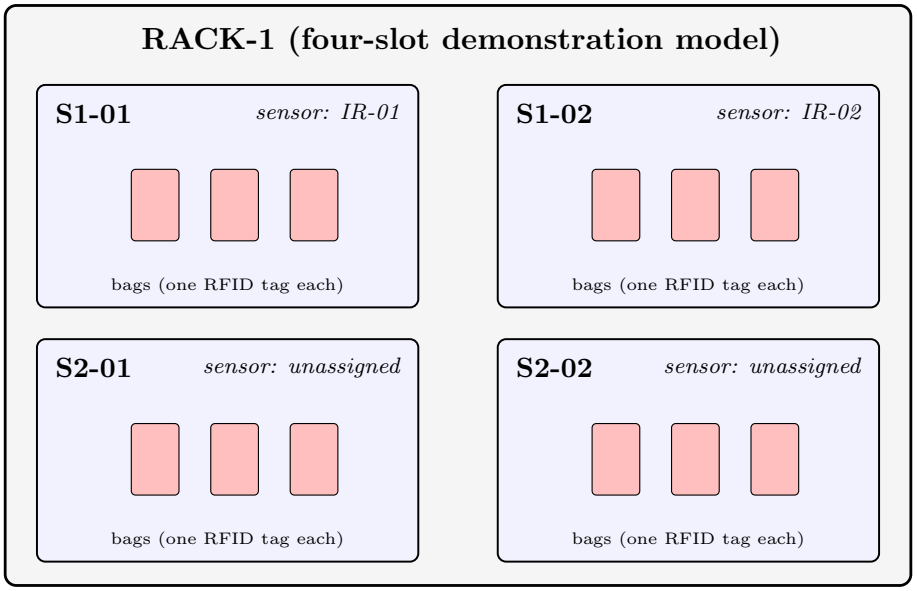


Figure 3.5: RFID-to-Database Data Flow of a Blood-Bag Scan

Four-Slot Storage Model

The four-slot storage model of the rack is shown in Fig. 3.6. The rack is identified as `RACK-1`. Its slots are `S1-01` and `S1-02` in the first row and `S2-01` and `S2-02` in the second row. In the current configuration, infrared sensors IR-01 and IR-02 are assigned to the first two slots, and the remaining slots do not yet have a sensor assigned. Each slot is drawn with several bag positions to show that more than one bag can be associated with a slot, with every bag having its own RFID tag and record.



Blood group assigned to each slot is a rack configuration (not fixed by the design)

Figure 3.6: Four-Slot Storage Model with Multiple Bags per Slot (conceptual illustration)

Database Design

The database records the entities that appear on the dashboard. The known fields of the `public.devices` table are `id` (UUID), `device_code`, `device_name`, `status`, `last_seen_at`, and `created_at`. The `blood_bags` table holds one record per bag, and `blood_bags.created_at` is the agreed source of the time at which the bag entered the system. The dashboard also shows information about bag group, expiry date, slot location, status, temperature, storage slots (rack, slot, occupancy, sensor, update time), and activity events (event type, description, bag, time). Fig. 3.7 shows a conceptual relationship between these entities. It is a conceptual diagram drawn from the information presented by the dashboard and the known device fields. The complete column list of each table is defined in the project database.

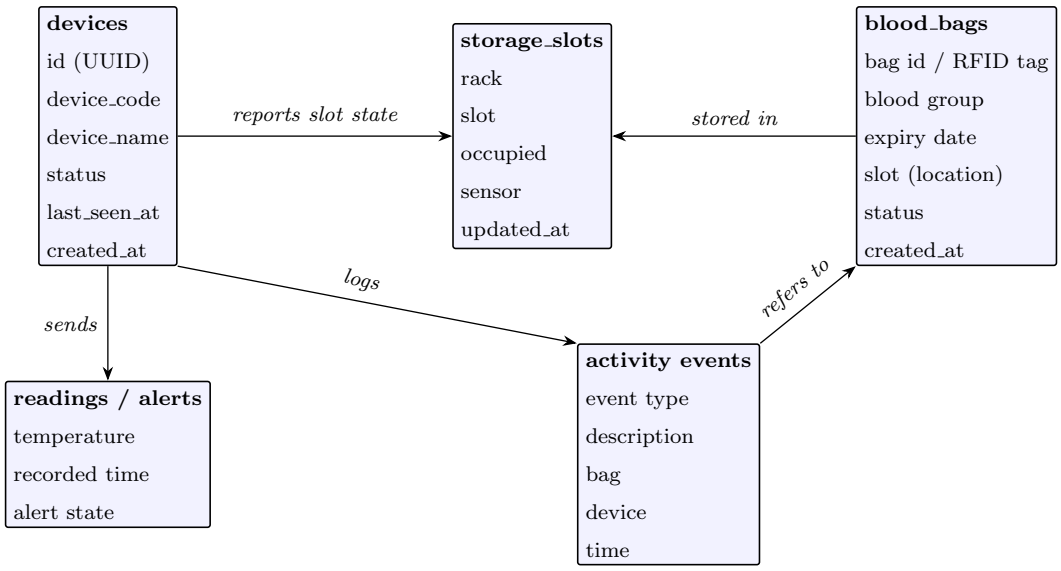


Figure 3.7: Conceptual Relationship Between the Main Data Entities (proposed; derived from dashboard views)

Security Considerations

Because the system handles information about blood inventory, secure design is important. The following measures describe the approach adopted for the design and the direction for hardening it.

- **Authentication:** Devices communicate with the backend through the Edge Function, which should accept only requests that carry valid device credentials. The operator signs in to the dashboard.
- **Least Privilege:** Each client should have only the permissions it needs. The dashboard needs read access to the inventory views, while the devices need only the ability to submit scans, readings, and heartbeats through the Edge Function.
- **Input Validation:** The Edge Function should validate the format and range of every incoming field (tag identifier, device code, temperature value) before it is written to the database.
- **Row Level Security:** Row Level Security policies on the database tables restrict which rows each role can read or write [11], so that access is limited even when a request reaches the database.
- **Secret Management:** Secret keys and device credentials are never placed in the dashboard code, in the report, or in public repositories; they are stored as environment variables or secrets of the respective platform.

The prototype demonstrates the data flow; the full production-grade configuration of authentication and Row Level Security is listed under future work in Chapter 7.

CHAPTER 4

WORK PROGRESS AND RESULTS

4.1 Hardware Prototype

The hardware prototype was assembled as a 3D-printed rack with an integrated control panel, as shown in Fig. 3.1. The RC522 RFID reader is mounted at the lower left of the front panel, and a key-fob RFID tag can be presented to it. The OLED display sits in a window above the reader and shows a status message for the operator. The right-hand part of the front contains the compartments, which are closed by hinged doors with handles. Ventilation openings are provided at the bottom, and the open top of the enclosure gives access to the wiring and the ESP8266 controllers.

The ESP8266 nodes, the RFID reader, the OLED display, and the sensors were interconnected as described in Section 3.6, and a stable power supply was provided to all modules. After assembly, the prototype was powered and its modules were verified functionally through the dashboard outputs presented in Section 4.4.

The prototype presented here is the realisation of the concept proposed at the Phase 1 review. The Phase 1 proposal described a six-slot rack controlled by an ESP32 with additional LED and buzzer indicators. The prototype built for this phase is a four-slot demonstration model controlled by ESP8266 NodeMCU nodes, which keeps the hardware simpler and the cost low while still demonstrating every stage of the data pipeline. The architecture is not limited to this size.

4.2 Software Development Phase

The software development phase was carried out in the following stages.

Firmware Development: The firmware for the ESP8266 nodes was written to initialise the peripherals, connect to Wi-Fi, read RFID tags and sensor values,

update the OLED display, and send data to the backend. The RFID node and the sensor node run separate programs, and each identifies itself with its device code.

Backend Development: The Supabase project was created and the database tables for devices, bags, slots, readings, and activity events were set up. The Edge Function was developed to receive data from the devices, perform the server-side operations, and record heartbeats.

Dashboard Development: The React dashboard was developed with pages for the overview, inventory, storage, alerts, activity log, connectivity, and settings, and it was deployed on Vercel. It supports searching the inventory by bag, group, or slot and refreshing the data.

Integration: The firmware, the Edge Function, the database, and the dashboard were integrated, and the complete path from a scan to the dashboard was exercised repeatedly with the demonstration tags.

Table 4.1 separates the functions that have been implemented and observed from those that are in progress or proposed.

Table 4.1: Implementation Status of the Main Functions

Function	**Status**	**Evidence**
RFID scan of a registered bag and activity logging	Implemented	Activity log (Fig. 4.4)
OLED status display on the rack	Implemented	Prototype photographs
Device heartbeat and connectivity page	Implemented	Fig. 4.3
Slot occupancy via IR sensors (two slots assigned)	Implemented	Fig. 4.2
Temperature reading uploaded and shown	Implemented	Fig. 4.1
Inventory table with group, location, status, temperature	Implemented	Fig. 4.1
IR sensors for the remaining two slots	In progress	Slots shown as unassigned
Automatic expiry alerts (bags near expiry)	Proposed	Dashboard shows an expiry indicator; expiry data to be entered
Temperature-threshold alerts	Proposed	Threshold to be configured
Production-grade authentication and RLS policies	Proposed	Section 3.7
Multi-bank deployment	Proposed	Future scope

4.3 Results and Testing

The prototype was tested by using the demonstration RFID tags and observing the behaviour of the rack and the dashboard. The tests that were carried out and the tests that remain to be carried out are listed in Table 4.2. No quantitative accuracy or response-time figures are claimed, because the tests carried out were functional demonstrations and not measured trials.

Table 4.2: Test Cases and Their Status

No.	Test case	Expected behaviour	Status
1	Valid tag scan	Scan is logged for the bag and shown in the activity log	Observed (five scan events of bag BB01)
2	Repeated scan of the same tag	Each scan is logged as a separate time-stamped event	Observed (BB01 scanned on 11 Oct and 13 Oct)
3	Device heartbeat	Last-seen time of each node is recorded	Observed (both nodes)
4	Inactive device	Dashboard reports no recent heartbeat and counts active devices	Observed (0 of 2 active when nodes were off)
5	Slot occupancy update	Slot record shows occupancy and update time	Observed (S1-01, S1-02 updated)
6	Temperature reading	Reading uploaded and shown on overview and inventory	Observed (31.8 °C)
7	Dashboard update	Dashboard reflects database content	Observed
8	OLED feedback	Display shows system status	Observed (photograph)
9	Unknown (unregistered) tag	Scan reported as unregistered; no bag changed	Planned
10	Network failure	Node recovers and resumes after Wi-Fi loss	Planned
11	Multiple bags in one slot	Each bag has its own record and slot association	Planned
12	Out-of-range temperature	Alert raised when threshold is crossed	Planned

The key observations from the tests carried out are as follows.

- The RFID reader identified the tag of blood bag BB01, and each scan was recorded as an `rfid_scan` event by the device ESP8266-RFID-01 with the bag and the time of the scan.
- Repeated scans of the same tag, made at different times on 11 October and 13 October, were stored as separate events, which shows that the activity log preserves a history of scans.

- Both ESP8266 nodes registered heartbeats, and the connectivity page reported their last-seen times. When the nodes were not transmitting, the page reported "No recent heartbeat", and the sidebar showed "0/2 devices active", so inactive devices are made visible to the operator.
- The storage page listed the four slots of RACK-1 with their occupancy, their assigned sensors (IR-01 and IR-02 for the first two slots), and the time of the last update.
- The overview page showed one registered bag, one available bag, no bags expiring within seven days, the latest temperature, and no open alerts.
- The temperature shown, 31.8 °C, is an ambient reading of the demonstration environment taken with the prototype outside any refrigerated storage. It demonstrates that readings are captured and displayed, and it is not a reading of a refrigerated blood-storage condition.

4.4 Output Results

The outputs of the dashboard obtained during testing are shown in Fig. 4.1 to Fig. 4.4.

The **Overview** page in Fig. 4.1 gives the operator a summary of the stock. Its summary cards show the number of registered bags, the number of available bags (not issued or expired), the number of bags expiring within seven days, and the latest temperature with the time at which it was recorded. Below the cards, the blood-inventory table lists the bag identifier, its RFID marker, the blood group (O+ for bag BB01), the expiry date, the location (RACK-1, S1-01), the status (available), and the temperature. The storage-slot panel shows the four slots with the bag assigned to each, and the lower part of the page presents the recent activity and the open alerts. The sidebar shows the number of active devices.

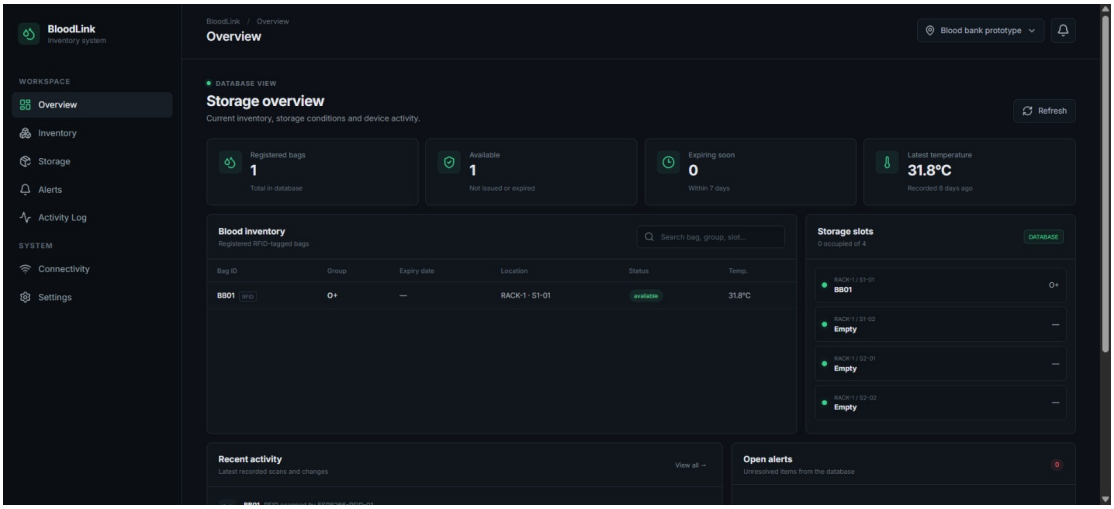


Figure 4.1: Dashboard Overview: Storage Overview with Inventory, Storage Slots, and Latest Temperature

The **Storage** page in Fig. 4.2 lists the configured rack slots and their occupancy. For every slot it shows the rack, the slot identifier, whether the slot is occupied, the assigned sensor, and the time of the last update. Slots S1-01 and S1-02 have the sensors IR-01 and IR-02, and S2-01 and S2-02 are unassigned. The occupancy of a slot is the live state reported by its sensor, whereas the location of a bag is the slot with which the bag is registered, so the two items of information can differ when a device is not transmitting.

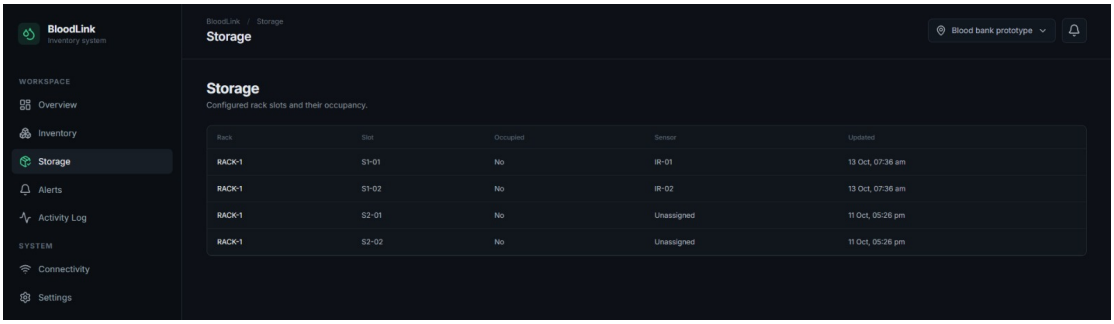


Figure 4.2: Dashboard Storage Page: Rack Slots, Occupancy, Assigned Sensors, and Update Time

The **Connectivity** page in Fig. 4.3 reports the state of each device based on the last heartbeat recorded by the Edge Function. Both nodes appear with their last-seen times. The screenshot was taken when the nodes were not transmitting, and the page correctly reported that there was no recent heartbeat.

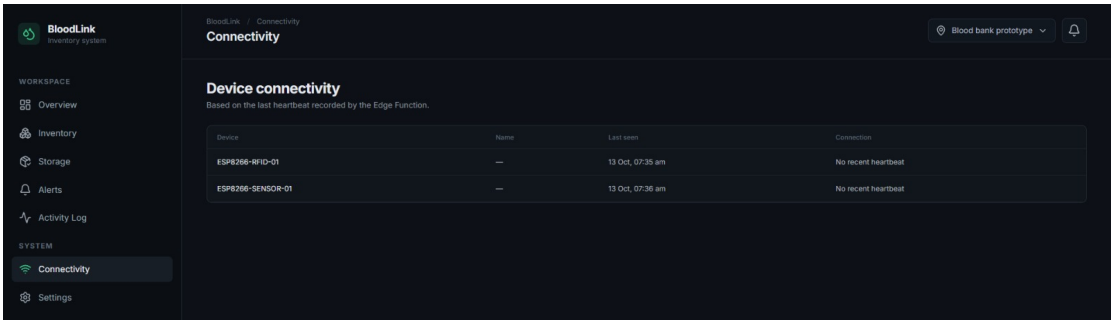


Figure 4.3: Dashboard Connectivity Page: Device Heartbeat Status

The **Activity Log** in Fig. 4.4 lists recent RFID scans and inventory events. Each row gives the event type, a description (for example, "RFID scanned by ESP8266-RFID-01"), the bag, and the time. The log is the basis for traceability, since it keeps a time-stamped history of how each bag was handled.

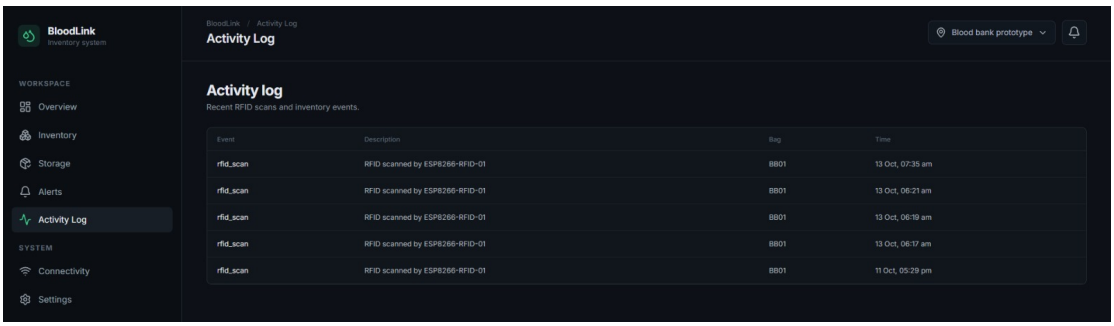


Figure 4.4: Dashboard Activity Log: Recent RFID Scans and Inventory Events

4.5 Cost Estimation

The estimated cost of the prototype is given in Table 4.3. The figures are the budget estimated by the team for the prototype. Actual component prices may vary with the supplier and the time of purchase.

Table 4.3: Cost Estimation of the Prototype

Sl. No.	**Category**	**Estimated Cost (Rs.)**
1	Electronics (ESP8266 nodes, RFID reader and tags, OLED, sensors, wiring, power supply)	1,600 – 2,600
2	3D printing of the rack and enclosure	500 – 1,500
3	Miscellaneous (connectors, fasteners, finishing material)	200 – 400
	Total Estimated Cost	**2,300 – 4,500**

The prototype was developed within an estimated total cost of Rs. 2,300 to Rs. 4,500. The software platforms used (React, Supabase, and Vercel) offer free tiers that are adequate for a prototype, so they do not add to the cost. The low cost shows that an RFID-based inventory and monitoring rack can be built with readily available components.

CHAPTER 5

CHALLENGES

The development of the BloodLink prototype involved several technical and practical challenges. They are described in two groups: those observed during the work and those that are anticipated for further development.

Challenges Observed During Development

1. **Enclosure Fabrication and Finishing:** The 3D-printed enclosure needed post-processing, such as sanding, patching, and fit adjustment of the panel openings, to make the display window, the reader opening, and the compartment doors fit properly.
2. **Hardware Integration and Cable Management:** Several modules (two ESP8266 boards, the RFID reader, the OLED display, the sensors, and the power supply) had to be fitted into a compact enclosure. Keeping the wiring tidy and the connections reliable inside the electronics bay required careful routing.
3. **Network Connectivity and Device Visibility:** The nodes depend on Wi-Fi, and a node that is off or out of range simply stops reporting. A heartbeat-based connectivity page was therefore needed so that the operator can tell the difference between a quiet device and a failed one.
4. **Consistency Between Bag Location and Slot Occupancy:** The registered slot of a bag and the occupancy reported by the sensor are separate pieces of information, and they can disagree when a device is not transmitting. The dashboard has to present both clearly.

Anticipated Challenges

5. **RFID Identification Reliability:** Read range, tag orientation, and reading several tags in a short time may affect identification when many bags are handled.
6. **Backend Communication and Database Consistency:** Lost requests, duplicate scans, and simultaneous updates must be handled so that inventory records stay consistent.

7. **Multi-Bag Inventory:** Handling many bags in the same slot requires reliable association between each tag, bag record, and slot.

8. **Sensor Calibration and Placement:** Temperature sensors need calibration, and their position decides what the reading represents. A reading of the ambient storage temperature is not a measurement of the temperature inside each bag.

9. **Power Management:** Continuous operation of both nodes needs a stable supply, and a loss of power should be detected and reported.

10. **Security:** Protecting device credentials, validating incoming data, and applying least-privilege access and Row Level Security are necessary before the system is used with real data.

11. **Data Accuracy and Validation:** Blood group, expiry date, and similar details depend on the correct registration of each bag by authorised staff, since RFID only identifies the tag.

CHAPTER 6

MEMBER ROLES

The responsibilities of the team members during the project are given in Table 6.1.

Table 6.1: Member Roles

Member	**Name (Roll No.)**	**Responsibilities**
Member 1	Abhijith Santhosh (MGP23ECE001)	Literature review, project documentation, 3D printing of the rack, hardware integration, and OLED display.
Member 2	Aswin S (MGP23ECE018)	ESP8266 firmware development and interfacing of the RFID reader.
Member 3	Elias Janson (MGP23ECE022)	Development and testing of the sensor node (IR slot sensors and temperature sensor).
Member 4	Johan Paul (MGP23CS082)	Supabase backend and Edge Function, database design, and development and deployment of the web dashboard.

All members took part in the system architecture discussions, in the integration of the hardware with the cloud and the dashboard, and in the final testing and demonstration.

CHAPTER 7

CONCLUSION

The project achieved the design and implementation of BloodLink, a prototype smart blood storage rack that combines RFID identification, slot and environmental sensing, cloud storage, and a web dashboard. The rack demonstrates the complete data path of the concept. A blood-bag tag is scanned by an RC522 reader connected to an ESP8266 node, the scan is passed to a Supabase Edge Function and recorded in the database, the OLED display gives local feedback, and the dashboard presents the inventory, storage slots, activity, and device connectivity.

The four-slot model shows how inventory can be organised by blood group, with several individually identified bags in each slot. The two RFID tags used in the demonstration illustrate the workflow, and the database model is not limited to two bags. The second ESP8266 node reports slot occupancy and temperature, and heartbeats make it clear to the operator which devices are active. The dashboard outputs obtained during testing confirm that scans, device status, slot status, and temperature readings are stored and displayed correctly.

A key advantage of the system is that it joins inventory tracking and storage monitoring, which are usually handled separately, in a low-cost design. The estimated cost of Rs. 2,300 to Rs. 4,500 and the use of open and free-tier platforms make the approach accessible to students and small facilities.

The limitations of the prototype must be stated clearly. BloodLink is a proof of concept and not a certified medical device. RFID identifies the tag and not the medical suitability of the bag, so the bag details depend on accurate registration and validation. The temperature readings are environmental readings, and the prototype has no refrigeration and does not show that blood is safe for transfusion. The acceptable temperature range is a configurable parameter that has to be set from the storage requirements of the blood component. The prototype has not been clinically validated, has no regulatory approval, and has not been deployed in a real blood bank.

Future Scope: The work can be extended in the following directions:

- Configurable temperature thresholds based on component-specific storage requirements, with automatic alerts for abnormal temperature.
- Automatic expiry alerts, using validated expiry dates entered at registration.
- Infrared sensors for all slots, a larger rack, and integration with a refrigerated storage unit.
- Production-grade authentication, least-privilege access, input validation, and Row Level Security.
- Handling of unregistered tags, duplicate scans, and network failures, with retry and offline buffering on the device.
- A mobile application, notifications, and reports for blood bank staff.
- Support for several blood banks, and formal testing with measured accuracy and response time.

Overall, the project shows how embedded systems, RFID, and cloud technology can be combined to give blood banks better visibility of their stock and storage conditions, and it provides a foundation for further development of a safe, traceable, and affordable blood inventory system.

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